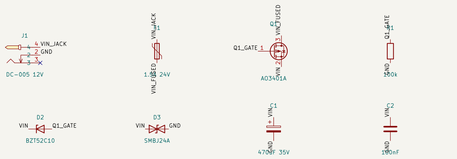
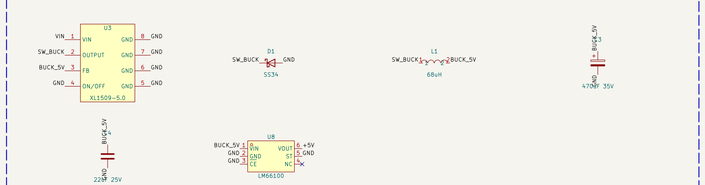
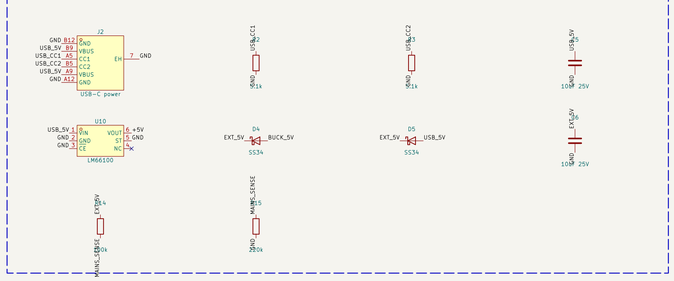
12 V input: fuse, reverse-polarity and surge protection



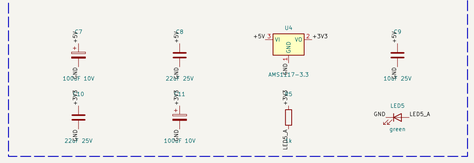
5 V buck converter (XL1509-5.0, 150 kHz)



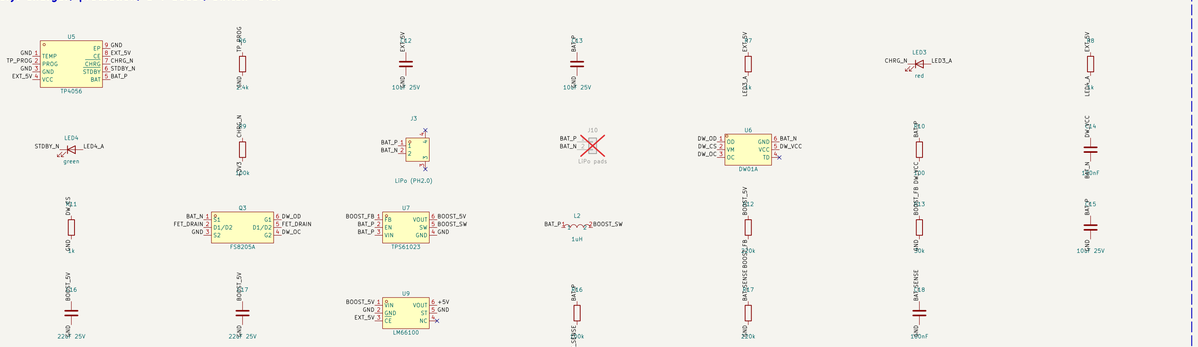
USB-C 5 V input (power only) and external-power sense



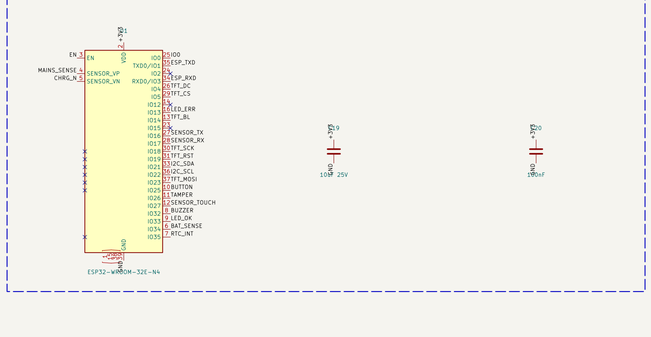
5 V rail and 3.3 V regulator



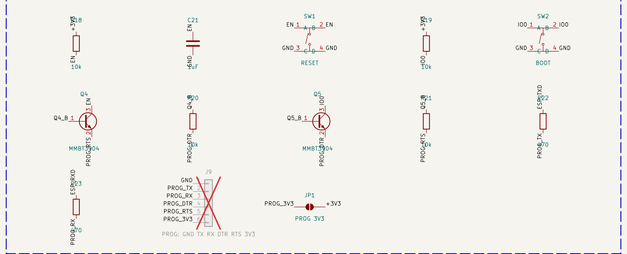
Battery: charger, protection, 5 V boost, switch-over



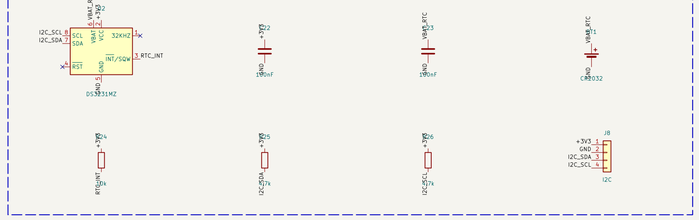
ESP32-WROOM-32E



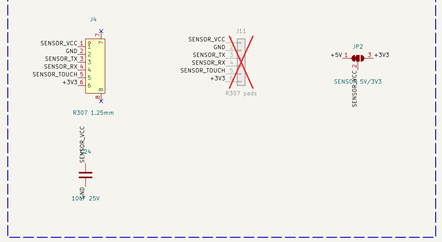
Reset, boot and programming header (no USB on the board)



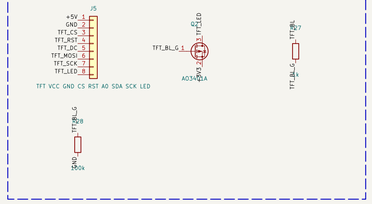
Real-time clock (DS3231M) and I2C



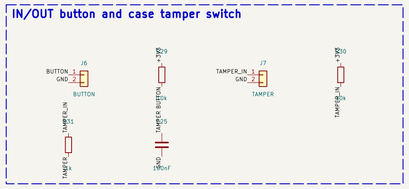
Fingerprint sensor (R307 / AS608)



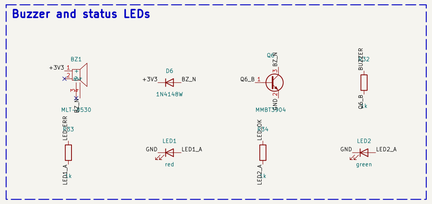
1.8" TFT (ST7735) with switchable backlight



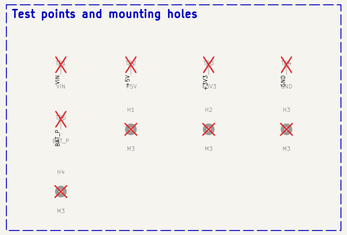
IN/OUT button and case tamper switch



Buzzer and status LEDs



Test points and mounting holes



Power flags: tell ERC these nets are supplied (connector, battery or regulator through passive pins)



Generated from pcb/generator/design.py — edit that file, not this schematic

Sheet: /	
File: attendance-board.kicad_sch	
Title: Attendance scanner main board	
Size: A1	Date: 2026-11-03
KiCad E.D.A. 10.0.7	Rev: 0.1
	Id: 1/1